

Program : Diploma in Chemical Engineering	
Course Code : 6072A	Course Title: Sustainable Energy Technologies
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To impart the importance of renewable energy resources and their utilization.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Environmental Science		Environmental Science	2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize present and future energy scenario of the world.	12	Understanding
CO2	Summarize various wind energy systems and solar energy harvesting methods.	18	Understanding
CO3	Summarize the methods for Bio energy generation from Bio waste.	18	Understanding
CO4	Summarise the tidal energy and hydrogen storage systems	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize present and future energy scenario of the world.		
M1.01	Outline the World Energy Usage	3	Understanding
M1.02	Summarise the environmental aspects of energy utilization	3	Understanding
M1.03	Summarise the Renewable Energy Scenario	3	Understanding
M1.04	Summarise the applications and economics of renewable energy systems	3	Understanding
M1.05	List various methods of energy sources	4	Understanding
Contents: World Energy Use- Reserves of Energy Resources Environmental Aspects of energy Utilization Renewable Energy Scenario in India and around the World- Potentials- Achievements Applications - Economics of renewable energy systems. Various methods of energy sources- Solar energy - wind energy- Bio-energy - Tidal power - fuel cells.			
CO2	Summarize various wind energy systems and solar energy harvesting methods.		
M2.01	Summarize solar energy Applications	4	Understanding
M2.02	Summarize Solar PV Power Generation and Applications	3	Understanding
M2.03	Identify various wind energy systems.	4	Understanding
M2.04	Summarize Site Selection, Safety and Environmental Aspects.	3	Understanding
	Series Test – I		

Contents:

Solar energy-Solar Radiation –solar constant- Measurements of Solar Radiation- Flat Plate and Concentrating Collectors- Solar direct Thermal Applications- Solar thermal Power Generation

Fundamentals of Solar Photo Voltaic Conversion- Solar Cells- Solar PV Power Generation- Solar PV Applications.

Wind Energy- Wind Data and Energy Estimation- Types of Wind Energy Systems- Details of Wind Turbine Generators.

Performance Site Selection- Safety and Environmental Aspects.

CO3	Summarize the methods for Bio energy generation from Bio waste.		
M3.01	Summarize Bio-Energy and biomass	5	Understanding
M3.02	Outline the various biogas plants and digesters for ethanol production	5	Understanding
M3.03	Summarize the cogeneration and utilization of Bio diesel	4	Understanding
M3.04	Identify the applications of biomass	4	Understanding

Contents:

Bio-Energy- definition- Biomass – Physical and chemical composition – properties of biomass

Biomass direct combustion.

Biomass Gasifiers- site selection of Biogas plants-material requirement for construction of Bio gas Plant- utilization –economics Digesters - anaerobic and aerobic combustion - Ethanol production.

Bio diesel-Cogeneration- utilization of Bio diesel – Technology for production of bio diesel - Transesterification – Process – Usage of Methanol – Glycerin – Storage and Characterization of biodiesel.

Biomass Applications- Bio fuels-bio diesel- bio ethanol and value addition of byproducts.

CO4	Summarize the tidal energy and hydrogen storage systems		
M4.01	Summarize other renewable resources	3	Understanding
M4.02	Summarize the tidal energy systems	3	Understanding
M4.03	Outline Small Hydro-Geothermal Energy	3	Understanding
M4.04	Summarize the various hydrogen storage systems	3	Understanding
	Series Test – II		

Contents:

Other renewable energy sources – Identify suitable energy sources for a location

Introduction - Tidal energy- Wave Energy- Open and Closed OTEC Cycles.

Illustrate Small Hydro-Geothermal Energy

Hydrogen and Storage- Fuel Cell Systems- advantages, Hybrid Systems.

Text / Reference

T/R	Book Title/Author
T1	Non-Conventional Energy Sources; Rai. G.D
R1	Renewable Energy Sources; Twidell, J.W. & Weir, A
R2	Solar Energy; Sukhatme. S.P
R3	Fundamentals of Renewable Energy Sources; GN Tiwari and MK Ghoshal
R4	Renewable Energy and Environment-A Policy Analysis for India, NH Ravindranath, UK Rao, B Natarajan, P Monga

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103206/